

NANYANG TECHNOLOGICAL UNIVERSITY
School of Mechanical and Aerospace Engineering

**EXPERIMENT E3.3: COMPUTER MODELLING, SIMULATION AND CONTROL OF
AN ELECTROMECHANICAL SYSTEM**

LOG SHEET

Name: _____ Matr.No.: _____

Date: _____ Time: _____ (AM / PM)

Submitted to: _____

6.1 a): _____

6.1 b): _____

6.2 a):

Type of response	k	Mp	tp	ts	error (%)
Under damped					2
Critically damped		n/a	n/a		2
Over damped		n/a	n/a		2

6.2 b): _____

7):

(Use additional pages, if needed)

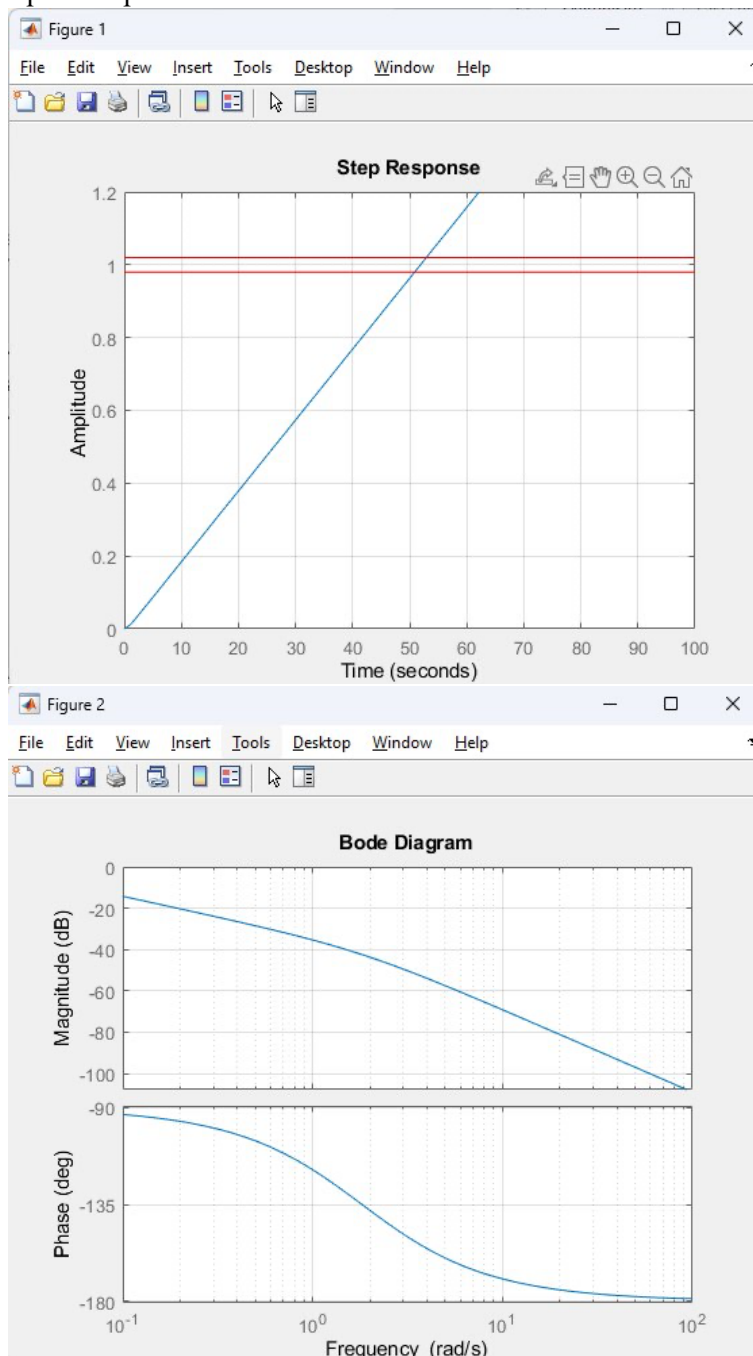
Plots

Using a screen capture tool (e.g. “Snipping Tool” in Windows), paste the plots related to:

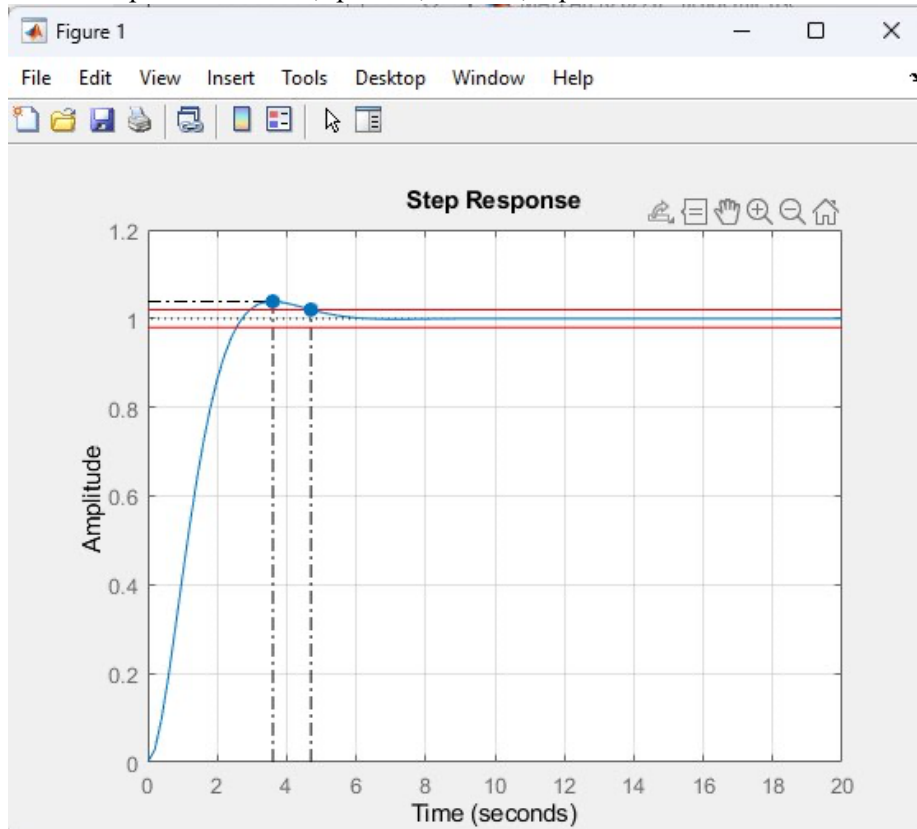
1. Open-loop step response
2. Open-loop Bode plot
3. Closed-loop step response for underdamped condition
4. Closed-loop step response for critically damped condition
5. Closed-loop step response for overdamped condition

You can paste two plots per page.

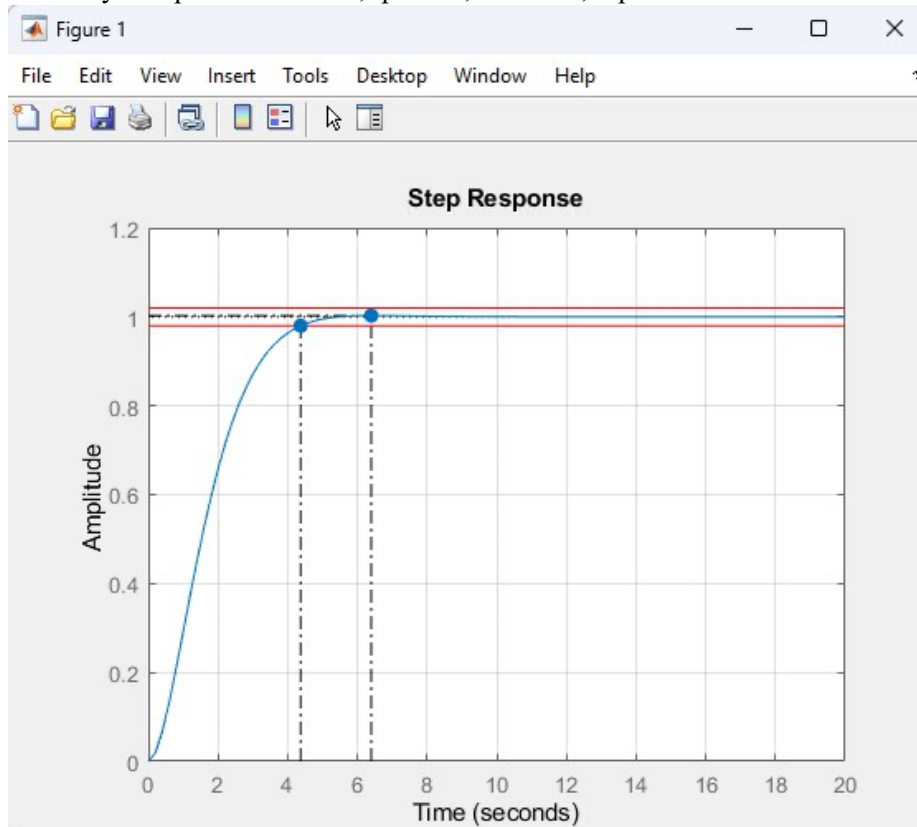
Open Loop with $k = 1$:



Underdamped with $k = 45$, $t_p = 3.6$, $t_s = 4.7$, $m_p = 3.85$:



Critically damped with $k = 30$, $t_p = 6.4$, $t_s = 4.37$, $m_p = 0.285$:



Overdamped with $k = 20$, $t_p = 20$, $t_s = 7.98$, $m_p = 0$:

